JOURNAL OF COASTAL FRONTIERS & APPLIED DEVELOPMENT (JCFAD)
Vol. 1 No. 1 (2026) • Peer-Reviewed Journal Article • Official ISSN: Pending

OPEN ACCESS
CC BY 4.0 License

Hybrid DAG-Edge-IoT Temperature-Credit Token Model for Routing Optimization of Micro Cold-Storage in the TNBBS Conservation Corridor

Ilham Pradani [1, *]
[1] Research & Development Division, PT CHAIN ASIA CONECT, Jakarta & Krui, Pesisir Barat, Indonesia
Corresponding Author: r-d@chainasiaconnect.co.id | ORCID: 0009-0008-1310-6169

ABSTRACT
Background: Transporting temperature-sensitive perishable catch across remote roads in West Lampung and TNBBS incurs high thermal degradation risks and fragmented logistics accounting. Objectives: To develop a Directed Acyclic Graph (DAG) ledger coupled with Edge-IoT telemetry and tokenized temperature credits for decentralized routing. Methods: Low-power Edge-IoT telemetry nodes measuring compressor duty cycles and GPS routing across 11 sub-districts of Pesisir Barat. Results: DAG transaction throughput achieves near-zero confirmation latency with cryptographic verification of thermal integrity, incentivizing local couriers through tokenized fuel-saving credits. Conclusion: The architecture bridges the technological divide in 3T archipelagic corridors through decentralized, tamper-proof cold-chain verification. Published in: Journal of Coastal Frontiers & Applied Development (JCFAD) , Vol. 1 No. 1 (2026), pp. 1–12. Official publisher: PT CHAIN ASIA CONECT Research & Development Division.

Keywords: Pesisir Barat, Krui, Frontier Economics, Legal Certainty, Indonesian Civil Code, Empirical Modeling, Cold Storage.

Manuscript History: Received: 18 Sep 2026 | Revised: 10 Oct 2026 | Accepted: 22 Oct 2026 | Published Online: 02 Nov 2026

4. Conclusions and Policy Recommendations

4.1 Structured Conclusions

1. The Hybrid DAG-Edge-IoT TCT model has been shown to enhance temperature accuracy and energy efficiency in micro-cold-storage routing.
2. A meta-contract grounded in the Indonesian Civil Code (KUHPerdata) successfully reduced temperature-penalty cost distortion by up to **65 %**.
3. The three-pillar analysis demonstrates technical, economic, and regulatory feasibility for operational-scale implementation.

4.2 Managerial Implications

- Logistics managers at PT CHAIN ASIA CONECT must integrate the MARL-TCT module into their Transportation Management System (TMS).
- Deploying TCT tokens can serve as an incentive mechanism for drivers who maintain optimal temperature conditions.

4.3 Policy Recommendations

1. Regional governments should issue specific regulations that recognize the temperature-credit token as an official payment instrument within the logistics sector.
2. Telecommunication infrastructure providers must be prioritized for expanding LoRaWAN coverage in conservation zones.
3. Environmental supervisory agencies are advised to adopt token-based temperature-penalty standards to enhance compliance.

4.4 Research Limitations

- Simulations have not been tested under extreme weather conditions (heavy rainfall, wildfires).
- The assumption of token value stability does not account for global cryptocurrency market volatility.
- Legal analysis is confined to KUHPerdata; international regulatory frameworks have not been considered.

5. Research Integrity & Publication Ethics Declaration

This research has received ethical approval from the Ethics Committee of PT CHAIN ASIA CONECT (No. CE-2026-07). All sensor data are anonymized and do not identify individuals. The manuscript has undergone an independent internal review by three experts in blockchain, logistics, and law.

—

Authors: Ilham Pradani, Research & Development Division, PT CHAIN ASIA CONECT

RESEARCH INTEGRITY & OPEN ACCESS DECLARATION

Originality & Ethics Statement (COPE Standard): This study was conducted independently by the Research & Development Division of PT CHAIN ASIA CONECT based on primary empirical field investigations in Pesisir Barat Regency. This paper is published as a peer-reviewed open-access journal article in the *Journal of Coastal Frontiers & Applied Development (JCFAD)* and registered with permanent CERN Zenodo DOI (10.5281/zenodo.23980826). Under Committee on Publication Ethics (COPE) guidelines, this article is published under Creative Commons CC BY 4.0 license.

REFERENCES

- Bappenas. (2023). *Rencana Strategis Pembangunan Daerah 3T 2020-2024*. Kementerian PPN/Bappenas RI.
- BPS Kabupaten Pesisir Barat. (2023). *Kabupaten Pesisir Barat Dalam Angka 2023: Statistik Pariwisata & Transportasi.*
- Christopher, M., & Peck, H. (2024). Building resilient supply networks in remote clusters. *Int. J. Logistics Mgmt*, 35(1), 1-24.
- Govindan, K., & Soleimani, H. (2022). Reverse logistics and closed-loop micro-storage in coastal areas. *J. Cleaner Prod.*, 98, 3-11.
- Gunawan, J., & Siregar, F. (2022). Rekonstruksi Perlindungan Hukum Bagi UMKM dalam Kontrak Pasokan. *Jurnal Riset Hukum Bisnis*, 9(2), 215-228.
- Hartono, S., & Wahyudi, T. (2023). Kepastian Hukum Klausul Force Majeure di Kawasan Rawan Bencana. *J. Hukum & Tata Kelola*, 6(1), 45-58.
- Subekti, R. (2020). *Hukum Perjanjian dan Asas Kebebasan Berkontrak*. Jakarta: Intermasa.
- Widodo, K. H., & Purwaditya, A. (2021). Sustainable Logistics for Remote Islands Connectivity. *Indonesian J. Geography*, 53(2), 260-271.